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RYAN F. JOHNSON, PhD

Scientist & Engineer

EMPLOYMENT AND POSITION HISTORY

VISITING SCHOLAR **Sep 2024 - Present**

Stanford University, Employer Naval Research Laboratory (see below)

- Develop scalable and robust code-generator, ChemGen, for complex chemical kinetics for CFD
- Investigate accelerated algorithms for computational fluid dynamics, with focus on AI-based strategies
- Minimize uncertainty in foundational fuel chemistry model, improving reaction rates.
- Transition neural nets for chemical state evolution in CFD, optimizing time-split chemistry.
- Assisted in developing machine-learned linear Fourier neural operators for predicting fuel injection in supersonic cross flow
- Advise students on computational physics, facilitating GPU algorithm transitions for their work, assisting in proposal writing, teaching about computational fluid dynamics

SENIOR SCIENTIST AND AEROSPACE ENGINEER **Feb 2015 - Present**

Naval Research Laboratory, code 6041, Laboratory for Computation Physics and Fluid Dynamics ***Washington, DC***

- Project Lead for High-Speed Propulsion and Combustion across multiple projects
- Lead software developer for the JENRE® Multiphysics Framework (over 1000 successful merge requests and closed issues)
- Expertise in implementing a wide array of combustion models (from finite rate kinetics to flamelet models) of various fluid dynamic fidelity (from DNS to LES/RANS) into different numerical frameworks (from finite difference to FV/FEM-DG)
- Developed essential stability tools for high-fidelity simulations of chemically reacting flows and has co-developed seminal work in structure-preserving methods, a focus he continues to actively pursue
- Expertise in Computational Fluid Dynamics, from meshing complicated geometries to detailed analysis of large data sets
- Extensive experience in project design and competitive proposal writing
- *Current research projects:* Discontinuous Galerkin finite element methods for hypersonic propulsion (principal investigator), Solid fuel combustion (NRL principal investigator on one of several projects, co-investigator on others), Scramjet/ramjets (principal investigator), Physics informed machine learning (PIML) for hypersonic applications including replacement of chemical kinetics with ANNs (co-investigator), The mixing of chemically reacting supercritical fuels with sub-critical air (principal investigator), Multi-material heat transfer (solid-gas) in presence of gaseous combustion (principal investigator), Rotating detonation engines (RDEs) and fundamentals of detonations (co-investigator)
- *Past research projects:* Flameless/distributed combustion (principal investigator), Hypersonic weather impact and droplet breakup (co-investigator), The role of endothermic fuels in the cooling of combustor walls (principal investigator), Modeling the transport of contaminants in canine test facilities (co-investigator)

GRADUATE RESEARCH ASSISTANT **May 2010 – Feb 2015**

University of Virginia, Mechanical and Aerospace Engineering Department ***Charlottesville, VA***

- Developed various CFD codes included optional sub-models for: finite rate chemistry, coupled heterogeneous interface reactions, and detailed transport
- Investigated carbon surface reactions, counterflow flames, tube reactors, and high-speed combustion.
- Assisted in experimental configurations for both canonical flame experiments and high-speed propulsion

EDUCATION

PH.D. IN MECHANICAL AND AEROSPACE ENGINEERING **May 2010 - Nov 2014**

University of Virginia, GPA 3.76 ***Charlottesville, VA***

Dissertation: Multidimensional Reacting Flow Simulation with Coupled Homogeneous and Heterogeneous Models

BACHELOR OF SCIENCE IN AEROSPACE ENGINEERING **Sep 2006 - May 2010**

University of Virginia GPA 3.56, Cum Laude ***Charlottesville, VA***

AWARDS

Presidential Early Career Award in Science and Engineering (PECASE) 2025

U.S. Naval Research Laboratory's Sabbatical Program 2023

"Disruptive Finite Rate Kinetics For Navy Relevant Fuels" Competitive sabbatical assignment to Stanford University

U.S. Naval Research Laboratory's Alan Berman Research Publication Awards 2023

"High-performance computing for hypersonic simulations: Progress and challenges." Journal of DoD Research and Engineering, cash award

U.S. Naval Research Laboratory's Alan Berman Research Publication Awards 2022

"A Total Energy Formulation of the Flamelet Progress Variable Using the Discontinuous Galerkin Method." JANNAF, cash award

U.S. Naval Research Laboratory's Alan Berman Research Publication Awards 2021

"A conservative discontinuous Galerkin discretization for the chemically reacting Navier-Stokes equations" Journal of Computational Physics, cash award

Jerome Karle's Fellowship, Naval Research Laboratory 2015-2017

Selective fellowship for direct-hire NRL employees, two years of research start-up funds with $25K for startup research (travel + equipment)

Graduate Teaching Award 2014

Given to graduate students who contribute above and beyond to teaching at UVA, cash award

Farragh Graduate Fellowship 2013

Applied after NDSEG expiration, served as instructor for Partial Differential Equations (see above), pays competitive stipend for semester

Virginia Space Grant Consortium Graduate Fellowship 2011 and 2012

Additional $5K in stipend per year

National Defense Science and Engineering Graduate Fellowship 2010

Awarded 2010, extension granted to summer of 2014

RESEARCH INTERESTS

Propulsion modeling: Focus on high-speed airbreathing devices such as solid-fueled ramjets (SFRJs), ramjets, scramjets, rotating and oblique detonation engines using scale-resolved CFD

Transition of AI to production level software: Embedded AI for CFD, Neural/tensor network-based CFD frameworks, ML preconditioners, ML chemistry integration

Advanced subsystem physics: solid fuel regression, plasma-assisted combustion, complex fuel injection, flame stabilization, off-design detonations, and sustainable aviation fuels.

Solver and code development for propulsion: hardware-accelerated solvers (including GPU-based), and developing fast implicit solvers for complex physics

Chemical kinetics integration and acceleration: Accessible code development for detailed kinetics and transport; advancing fast chemistry solvers and preconditioned integration strategies.

Structure-preserving and high-order numerical methods: Conservative methods for unstructured grids with focus on chemically reacting flows and complex equations of state

Reduced-order modeling and compression strategies: Progress variable state-reduction approaches, employing AI-based techniques including tensor netowrks and machine learning to compress models.

Emerging compression strategies: Fourier neural operators and tensor networks, real-time learning and acceleration in CFD simulations.

Humanitarian demining: Removing landmines through advanced detection strategies, predicting explosive off-gas transport in soil

Hybrid propulsion systems and electric propulsion (Exploratory): from hybrid rockets to electric propulsion systems, with application to sustainable and interplanetary travel.

Sustainable energy conversion systems (Exploratory): adapting my CFD and kinetics modeling methodologies to simulate ground-based power systems fueled by sustainable technologies.

The importance of civil service (Exploratory): How civil service has a positive effect on society

TEACHING EXPERIENCE

TEACHING EXPERIENCE FALL 2013 & FALL 2014

Instructor - Partial Differential Equations, University of Virginia Engineering Department

- Class focused on solving and analyzing multidimensional continuum-based physics problems
- 4.77/5.0 in 2013 and 4.88/5.0 in 2014 in teaching evaluation
- Completely restructured undergraduate class
- Fall 2014 term under full autonomy
- Used flipped method for teaching
- Added elements of intro to computation and data processing through Python and MATLAB modules
- Wrote textbook: Handbook of Partial Differential Equations (located here)

SKILLS

Proposal Writing: Since joining NRL in 2015, I quickly transitioned to developing independent research programs that fully funded my work. I have written successful proposals across a range of disciplines, primarily focused on computational physics and transport phenomena. These efforts have allowed me to build and lead my own research team while also providing partial support for their individual research directions. Additionally, I have curated collaborative research initiatives to complement my core programs, resulting in multi-agency (DoE and DoD) partnerships with LLNL, ANL, and APL, collectively securing approximately $1.5 million in annual funding.

Program Development: I have assisted in the development of five Multi-University Research Initiatives, serving as technical point of contact, co-writing the call for proposals, and served on selection committee. I have helped NRL build several accomplished multi-division research initiatives that brought together broad research expertise.

Mentoring: Since 2015, I have mentored over twenty students at NRL, tailoring experiences to their academic level and helping many transition to top graduate programs or careers in national labs and industry. As my role expanded, I also guided new hires—several of whom now lead projects and shape group strategy, contributing to my promotion to senior scientist in under ten years, with mentorship cited as a key factor. During my 2024–2025 sabbatical at Stanford, I continued mentoring graduate students in collaboration with Professors Hai Wang and Beverley McKeon, resulting in various research works and ongoing advisory relationships.

Technical skills: C++, Python, FORTRAN, MATLAB, Version Control (git), Linux, CFD Software: The JENRE Multiphysics Framework, PeleC, OpenFOAM, StarCCM, Project Management, Technical Writing, Data Analysis, Proposal Writing, Machine Learning, Combustion Modeling, Fluid Dynamics, Chemical Kinetics, Technology Transition, Reduced Order Modeling

MENTORSHIP

Below is a list of individuals I have mentored and advised throughout my professional career. These relationships span from my time as a Ph.D. candidate at the UVA, through mentoring interns and early-career professionals at the NRL, to advising graduate students during my sabbatical at Stanford University.

Deigo Destino Ortiz (expected undergraduate Stanford 2026) 2024-2025
Mentored under Prof Beverley McKeon during Stanford sabbatical
Federico Rios Tascon (expected PhD Stanford 2026) 2024-2025
Mentored under Prof Beverley McKeon during Stanford sabbatical
Ethan Genter (expected PhD Stanford 2026) 2024-2025
Mentored under Prof Hai Wang during Stanford sabbatical
Kevin Dong (expected PhD Stanford winter 2025) 2024-2025
Mentored under Prof Hai Wang during Stanford sabbatical
Hamil Patel (expected PhD UCF winter 2026) 2024-2025
Intern at NRL, transitioned back SDSU, return 2023
Jay Sampayan (surname change Arcities now PhD candidate) 2024-2025
Intern at NRL, co-advised with Dr. Eric Ching, transitioned back SDSU, returned in 2025
Juan Cruzz Pelligrini (now PhD at Spectre Aerospace) 2023
Co-advised intern with Dr. Eric Ching, returned to UCF PhD program
Jacklyn Higgs (now PhD at ARL) 2023
Co-advised intern with Dr. Eric Ching, returned to UCF PhD program
Sarah Burrows (now PhD candidate UCF) 2023
Intern at NRL, transitioned to graduate school at UF with NDSEG Fellowship
Eric Ching 2021
NRL new hire, I onboarded and helped developed professionally at NRL
Brian Bojko 2021
NRL new hire, I onboarded and helped developed professionally at NRL
Cal Rising (now professor University of Texas El Paso) 2021
Intern at NRL, transitioned to NRL full time
Andy Huynh (now PhD candidate Stanford) 2020
Co-advised shared intern with Dr, Brian Fisher, transitioned to graduate school at Stanford University
Del Irvin (now at Hermus) 2016
Co-advised shared intern with Professor Harsha Chelliah at UVA
Devon Wood-Thomas (now PhD Princeton) 2015
Co-advised at NRL internship, mathematics and geometry for CFD, transition: Princeton undergraduate
Deborah Hoffman Neumann (now at Aurora Flight Sciences) 2014
Co-advised design UVA senior thesis, nozzle design for counterflow experiments, transition: Rolls-Royce now

Alexandra VanDine (now PhD MIT Lincoln lab) 2014
Co-advised at UVA for ongoing research, mesh development for simulations, transition: UCSD PhD now

SERVICE

Below is a list service I have done in both my personal and professional life

Civil Service 2015-present
United States civil servant through the department of defense
AIAA member 2008-present
Member to professional society American Institute of Aeronautics and Astronautics
Combustion Institute Member 2025-present
Participant in combustion institute events, such as united states combustion meeting and proceedings of combustion institute
AIAA Session Chair 2015-present
Organize and assist in the operation technical presentation sessions at AIAA's Scitech
Combustion Institute Media & Outreach Committee 2025-present
Four year term member of the media outreach committee of the combustion institute
NREIP Intern Summer Presentation Judge 2015-present
Judge presentations and provide feedback to students on a three day event hosted by NRL following intern program
Reviewer: Combustion Theory and Modeling 2015-present
Reviewer: Combustion and Flame 2015-present
Reviewer: Journal of Propulsion and Power 2020-present
Reviewer: Proceedings of the Combustion Institute 2020-present
Reviewer: AIAA Journal 2020-present
Church Choir/Musician 2017-present
Helped arranged, compose, and perform as guitarist and singer in local catholic Christmas and easter masses
CTR^2 Creating Tomorrow's Research Program 2025
Participated in Q&A session on careers in government laboratory science with CTR's postdocs
CTR^2 Summer Outreach program 2025
Participated in Q&A session on careers in government laboratory science
AIAA Special Session Organizer 2024-2025
Organized special session on fuel jets in cross flow
Sofar Sounds, songs from a room 2017-2021
Volunteer to assist in small pop-up shows around the DC area for up-and-coming musicians 10-15 hours a week
Charity Music Shows at XII in Charlottesville, VA Oct 2018
Helped raise money and awareness
Career day information sessions for elementary students at Powell elementary school Oct 2016,17,18
Helped raise money and awareness
High school math tutor for Anacostia local high schools 2015-2019
Volunteered help with students local to NRL, program unfortunately cancelled

PROJECT MANAGEMENT EXPERIENCE

LISTED BELOW ARE PROJECTS FOR WHICH I HAVE SERVED OR CURRENTLY SERVE AS PRINCIPAL INVESTIGATOR.

CURRENT PROJECTS

Disruptive Reduced Finite Rate Kinetics for Navy-Relevant Fuels

Funding Source: Advanced Graduate Research Program, US Naval Research Laboratory
Funding Profile: $600K
Destination: Laboratory of Professor Hai Wang, Stanford University
Description: This project focuses on developing reduced finite rate kinetics (RFRK) models to improve the accuracy and efficiency of chemically reacting flow simulations for Navy propulsion systems. By integrating Hybrid Chemistry (HyChem) techniques and advanced computational methods, Dr. Johnson will refine fuel chemistry models to balance computational cost and fidelity. The research will enhance CFD frameworks and establish NRL as a leader in generating in-house RFRK models, reducing reliance on external sources.

Discontinuous Galerkin Methods for Modeling Chemically Reacting Hypersonic Phenomena

Funding Source: ONR 6.2 (PM: Dr. Eric Marineau)
Funding Profile: FY24 $255K, FY23 $355K, FY24 $395K, Total: $1.005 million
Collaborators: NRL is individually funded but we are connected to Argonne National Laboratory, Lawrence Livermore National Laboratory, Applied Physics Laboratory, through their own individual projects. The total project is close to 1.8 million a year.

Description: Focuses on high-order DG methods for large-scale chemically reacting hypersonic flows, time-integration techniques, computational optimization, and propulsion demonstration cases. Multi-agency collaboration.

PREVIOUS PROJECTS

Understanding the Impacts of Cooling Hypersonic Engine Surfaces with Jet Fuels

Funding Source: NRL Base 6.1 (Funds held at NRL that are competitively bid on internally)
Funding Profile: FY22 $360K, FY23 $360K, FY24 $360K, Total: $1.08 million
Description: Investigates computational modeling of endothermic pyrolysis of jet fuels under supercritical conditions for hypersonic engines. High-order simulation of chemically reacting fuel injection using GPU acceleration. Collaborative work with the NRL's 8200 Hypersonics Section.

High-Fidelity Simulations of Combustion in High-Speed Propulsion Engines

Funding Source: ONR 6.2 (PM: Dr. Eric Marineau)
Funding Profile: FY22 $200K, FY23 $200K, Total: $400K
Collaborators: NRL is individually funded but we are connected to Argonne National Laboratory, Lawrence Livermore National Laboratory, Applied Physics Laboratory, through their own individual projects. The total project is close to 1.8 million a year.
Description: High-fidelity simulations of combustion in high-speed propulsion engines. Goal of project was successful transition of the JENRE® Multiphysics Framework to our collaborators and assisting them with their research endeavors.

Investigating the Transport of Explosive Vapors

Funding Source: ONR 6.2 (PM: Dr. Joong Kim)
Funding Profile: FY22 $200K, FY23 $300, Total: $500K
Description: Focused on computational modeling of explosive vapor transport in various environments. The main application was understanding surface contamination from buried explosives that can be detected by Canines. Completed in CY23.

Stable Combustion and Efficient Thrust Generation in Hydrocarbon-Fueled Scramjets

Funding Source: NRL Base 6.2 (Funds held at NRL that are competitively bid on internally)
Funding Profile: FY21 $142K, FY22 $157K, FY23 $157K, Total: $456K
Description: Collaborative effort to develop high-fidelity scramjet combustion models beyond reduced-order techniques. Led to advances in predicting scramjet unstart phenomena.

High Speed Chemically Reacting Stagnating Jets

Funding Source: Jerome Karle Fellowship
Funding Profile: FY16 $220K, FY17 $230K
Description: Highly competitive entrance to NRL service fellowship that funds the first two years of federal service. Focus was on developing chemically reacting flow models into a GPU based code

INVITED TALKS AND PANELS

Guest Technical Talk: USC and UCI Joint Combustion Research Overview — Aug 2025
Joint talk between Prof Xu (USC) and Prof Shi (UCI)
Guest Lecture: Numerical Simulation of Chemically Reacting Flow — May 2025
Class: Stanford's ME372, Combustion Applications
Guest Lecture: Scale Resolved Simulations of Unsteady Chemically Reacting Flows — May 2025
Class: Stanford's ME451B, Flow Instabilities
Seminar: Stanford University Center for Turbulence Research Tea Seminar — May 2025
Host: Professors Beverley McKeon and Parviz Moin
Seminar: San Diego State University Mechanical Engineering Seminar — Apr 2025
Host: Professors Pavel Popov
Review Panel: National Science Foundation Combustion and Fire Portfolio — January 2025
Host: NSF PO Professor Harsha Chelliah
Seminar: Stanford University Thermofluids, Energy, and Propulsion Systems Group — October 2024
Host: Professor Hai Wang and Dr. Nick Kateris
Special Session: International Conference in Numerical Combustion — April 2024
Host: Professor Venkat Raman
Seminar: Georgia Institute of Technology Computer Science and Engineering — November 2023
Host: Professor Spencer Bryngelson
Panel: Priorities in Hypersonic Research — April 2023
Host: National Defense Science and Engineering Graduate Fellowship Research Conference
Panel: AIAA Careers in Propellants and Combustion — January 2023

Host: Professor Jacquline O'Connor
Seminar: University of Virginia Mechanical and Aerospace Engineering October 2021
Hosts: Professors Chloe Dedic and Harsha Chelliah
Seminar: University of South Carolina Aerospace Engineering September 2021
Host: Professor Sang Hee Won
Invited Talk: ONR/AFOSR Turbulence-Chemistry MURI Kickoff December 2020
Hosts: Professor Venkat Raman and Dr. Eric Marineau
Seminar: University of Arizona Aerospace Engineering April 2020
Host: Professor Kyle Hanquist
Review Panel: ONR/AFOSR Turbulence-Chemistry MURI April 2020
Host: Dr. Eric Marineau
Invited Talk: Multi-Agency Coordinating Committee on Combustion Research May 2019
Host: Dr. Chiping Li

JOURNAL PUBLICATIONS

Ryan F. Johnson, Eric J. Ching, Andrew D. Kercher, Jay Arcities, Joshua E. Lipman, Ethan S. Genter, Hai Wang "ChemGen: Chemistry Reaction Code Generation for Computational Physics" Computer Physics Communication, under review August, 2025
Contribution: Lead author, conceptualization, writing, and methodology

E. J. Ching, **R. F. Johnson,** "Effect of ozone sensitization on the reflection patterns and stabilization of standing detonation waves induced by curved ramps," In-press Aerospace Sciences and Technology, Sept 1 2025
Contribution: Resource management (PI of project), conceptualization, and methodology

Cal Rising, Eric J. Ching, **Ryan F. Johnson**,"Simulation of Non-Premixed, Supersonic Combustion using the Discontinuous Galerkin Method on Fully Unstructured Grids" Physics of Fluids, in preparation Aug 2025
Contribution: Resource management (PI of project), conceptualization, writing, and methodology

Jay Arcities, **Ryan F. Johnson**, Eric J. Ching, Kamal Viswanath, Joshua E. Lipman, Ethan S. Genter, Wedni Dong, Hai Wang "CodeJeNN: Imbedding Inference for Computational Physics" Software X, under review August, 2025
Contribution: Resource management (PI of project), mentorship of team, conceptualization, writing, and methodology

Yue Zhang, Wendi Dong, Nobilli Andrea, **Ryan F. Johnson,** Gregory P. Smith, Hai Wang, "Foundational Fuel Chemistry Model 2 — Can data assimilation yield useful insights in reaction rate constants?" Combustion and Flame, under review August, 2025
Contribution: Co-mentorship of team and writing

E. J. Ching, **R. F. Johnson**, S. Burrows, J. Higgs, A. D. Kercher, "Positivity-preserving and entropy-bounded discontinuous Galerkin method for the chemically reacting, compressible Navier-Stokes equations," Journal of Computational Physics, 113746, August 2025.
Contribution: Resource management (PI of project), mentorship of team, conceptualization, writing, and methodology

E. J. Ching**, R. F. Johnson**, A. D. Kercher, "Conservative, pressure-equilibrium-preserving discontinuous Galerkin method for compressible, multicomponent flows," arXiv preprint arXiv:2501.12532, 2025. Under review.
Contribution: Resource management (PI of project), mentorship of team members, conceptualization, and methodology

Ryan DeBoskey, David A. Kessler, Brian Bojko, **Ryan F. Johnson**, Andrew Kercher, Eric Ching, and Venkateswaran Narayanaswamy "Development and Evaluation of Reduced Kinetics Models for 1,3-Butadiene–Air Combustion" Journal of Propulsion and Power
Contribution: Methodology, mentorship of select team members, conceptualization, and model development

R. DeBoskey, C. Geipel, D. Kessler, B. Bojko, B. Fisher, **R. F. Johnson**, V. Narayanaswamy, “Numerical and experimental investigation of flame dynamics in opposed-flow solid fuel burner,” Combustion and Flame, vol. 273, p. 113960, March 2025, Elsevier.
Contribution: Methodology, mentorship of select team members, and model development

E. J. Ching, **R. F. Johnson**, “Conservative discontinuous Galerkin method for supercritical, real-fluid flows,” arXiv preprint arXiv:2410.13810, 2024.
Contribution: Resource management (PI of project), conceptualization, and methodology

E. J. Ching, **R. F. Johnson**, A. D. Kercher, “Positivity-preserving and entropy-bounded discontinuous Galerkin method for the chemically reacting, compressible Euler equations. Part I: One dimension,” Journal of Computational Physics, March 2023.
Contribution: Resource management (PI of project), conceptualization, writing, and methodology

E. J. Ching, **R. F. Johnson**, A. D. Kercher, “Positivity-preserving and entropy-bounded discontinuous Galerkin method for the chemically reacting, compressible Euler equations. Part II: Multiple dimensions,” Journal of Computational Physics, March 2023.
Contribution: Resource management (PI of project), conceptualization, writing, and methodology

Kessler, D.A., et al. (including **R. F. Johnson**) (2023). "High-performance computing for hypersonic simulations: Progress and challenges." Journal of DoD Research and Engineering, Vol. 6, pp. 65-77.

Geipel, C.M., Bojko, B.T., Pfutzner, C.J., Fisher, B.T., and **Johnson, R.F**., “Regression of solid polymer fuel strands in opposed-flow combustion with gaseous oxidizer” Sept 1st 2022, J. Proceedings of the Combustion Institute,
Contribution: Conceptualization, writing, and methodology

C.J. Rising, G.B. Goodwin, **R.F. Johnson,** D.A. Kessler, J. Sosa, M. Thornton, K.A. Ahmed, “Numerical investigation of turbulent flame-vortex interaction in premixed cavity stabilized flames,” Aug 1st 2022, Aerospace Science and Technology,
Contribution: Conceptualization, writing, and methodology

M.R. Thornton, A.J. Morales, D.M Smerina, C.J. Rising, J. Sosa, **R.F. Johnson**, D.A. Kessler, G.B. Goodwin, K.A. Ahmed, “Mean pressure gradient effects on the performance of cavity stabilized flames. Aerospace Science and Technology ” Aug 1st 2022, Aerospace Science and Technology
Contribution: Conceptualization, writing, and methodology

Johnson, Ryan F. and Kercher, Andrew D.; " A Conservative Discontinuous Galerkin Discretization for the Chemically Reacting Navier-Stokes Equations"; Journal of Computational Physics, 2020
Contribution: Lead author, conceptualization, writing, and methodology

Johnson, Ryan Frederick, Chelliah, Harsha Kumar; "Numerical simulation of two-dimensional flow over a heated carbon surface with coupled heterogeneous and homogeneous reactions"; Combustion Theory and Modelling, 2017
Contribution: Lead author, conceptualization, writing, methodology, and model development

Viswanath, Kamal, **Johnson, Ryan,** Corrigan, Andrew, Kailasanath, K, Mora, Pablo, Baier, Florian, and Gutmark, Ephraim; "Flow Statistics and Noise of Ideally Expanded Supersonic Rectangular and Circular Jets"; AIAA Journal, 2017
Contribution: Conceptualization, writing, methodology, and model development

Johnson, RF, VanDine, AC, Esposito, GL, Chelliah, HK; "On the Axisymmetric Counterflow Flame Simulations: Is There an Optimal Nozzle Diameter and Separation Distance to Apply Quasi One-Dimensional Theory?"; Combustion Science and Technology, 2015
Contribution: Lead author, conceptualization, writing, methodology, and model development

SELECT CONFERENCE PROCEEDINGS

E. J. Ching, **R. F. Johnson**, A. D. Kercher, "Conservative, pressure-based discontinuous Galerkin method for compressible, multicomponent flows," AIAA SCITECH 2025 Forum, 1296, 2025.

C. Rising, E. J. Ching, **R. F. Johnson,** "Comparison of hydrogen chemistry models in high-enthalpy, compressible, autoignition-dominated supersonic combustion," AIAA SCITECH 2025 Forum, 0948, 2025.

Demir, S., **Johnson, R.,** Bojko, B.T., Corrigan, A.T., Kessler, D.A., Guthrey, P., Burmark, J., and Irving, S., "Three-Dimensional Compressible Chemically Reacting Computational Fluid Dynamics With Tensor Trains," AIAA SciTech 2024 Forum, 2024.

DeBoskey, R., Kessler, D., **Johnson, R.F**., Bojko, B.T., Johnson, R.F., and Goodwin, G.B., "The Influence of Chemical Reaction Models on Combustion Dynamics in an Opposed-Flow Solid Fuel Burner," AIAA SciTech Forum, National Harbor, MD, Jan. 23-27, 2023.

Rising, C.J., Goodwin, G.B., **Johnson, R.F.,** Ching, E.J. (2024). "On the use of h and p-refinement for simulations of high enthalpy, compressible, chemically reacting flow using a discontinuous Galerkin approach." In AIAA Scitech 2024 Forum.

Ching, E. J., **Johnson, R.,** Burrows, S., Higgs, J. P., and Kercher, A. D, "Positivity-preserving and entropy-bounded discontinuous Galerkin method for the chemically reacting, compressible Navier-Stokes equations (IR-6043-22-49-U)," AIAA SciTech Forum, January 2023, National Harbor, MD, AIAA-2023-0664.
Contribution: Resource management (PI of project), mentorship of team, conceptualization, writing, and methodology

Sarah K. Burrows, **Ryan F. Johnson**, Andrew D. Kercher "Development of a Model Verification Test Case for Chemically Reacting Flows: Mixing Layer Interacting with an Impinging Oblique Shock" NRL/MR/6040-, U.S. Naval Research Laboratory, September 2022.
Contribution: Resource management (PI of project), mentorship of team, conceptualization, writing, and methodology

E. J. Ching, **R. F. Johnson**, A. D. Kercher, "Positivity-preserving and entropy-bounded discontinuous Galerkin method for the chemically reacting, compressible Euler equations. Part I: One dimension," NRL/MR/6040-, U.S. Naval Research Laboratory, September 2022.
Contribution: Resource management (PI of project), mentorship of team, conceptualization, writing, and methodology

E. J. Ching, **R. F. Johnson**, A. D. Kercher, "Positivity-preserving and entropy-bounded discontinuous Galerkin method for the chemically reacting, compressible Euler equations. Part II: Multiple dimensions," NRL/MR/6040-, U.S. Naval Research Laboratory, September 2022.
Contribution: Resource management (PI of project), mentorship of team, conceptualization, writing, and methodology

E. J. Ching, **R. F. Johnson**, S. K. Burrows, J. Higgs, A. D. Kercher, "Positivity-preserving and entropy-bounded discontinuous Galerkin method for the chemically reacting, compressible Navier-Stokes equations," NRL/MR/6040-, U.S. Naval Research Laboratory, September 2022.

Bojko, B.T., **Johnson, R.F.,** Geipel, C.M., Fisher, B.T., and Kessler, D.A, "Investigation of Hydroxyl-Terminated Polybutadiene Combustion and Regression With an Opposed Flow Burner,", NRL/MR/6040-, U.S. Naval Research Laboratory, September 2022.

Alisha Sharma, **Johnson, Ryan F.,** David Kessler, and Adam Moses; " Deep Learning for Scalable Chemical Kinetics"; AIAA Scitech Jan 2020

Andrew M. Hess, David A. Kessler, **Ryan F. Johnson** and Brian T. Bojko. "Towards a Simulation Method for Aluminum-Particle-Enhanced Solid Fuel Combustion Devices," AIAA 2021-3621. AIAA Propulsion and Energy 2021 Forum. August 2021.

Douglas A. Schwer and **Ryan F. Johnson.** "Performance of Centerbody-less Rotating Detonation Combustors for an RDE Concept," AIAA 2021-3651. AIAA Propulsion and Energy 2021 Forum. August 2021.

Evan W. Hyde, Gabriel B. Goodwin, **Ryan F. Johnson** and Tonghun Lee. "Ethylene Combustion in an Axisymmetric Mach 4.5 Cavity," AIAA 2021-1467. AIAA Scitech 2021 Forum. January 2021.

Gabriel B. Goodwin, Christian L. Bachman, **Ryan F. Johnson** and David A. Kessler. "Synthetic freestream turbulence generation at an inflow boundary condition,"AIAA 2021-1846. AIAA Scitech 2021 Forum. January 2021.

Johnson, R. F., Corrigan, A. T., Kercher, A., & Chelliah, H. K. Discontinuous-Galerkin Simulations of Premixed Ethylene-Air Combustion in a Cavity Combustor. In AIAA Scitech 2019 Forum, 2019

Douglas A. Schwer, **Ryan F. Johnson**, Andrew Kercher, David Kessler and Andrew Corrigan Progress in Efficient, High-Fidelity, Rotating Detonation Engine Simulations In AIAA Scitech 2019 Forum, 2019

Douglas A. Schwer, **Ryan F. Johnson**, Andrew Kercher, David Kessler and Andrew Corrigan "Progress in Efficient, High-Fidelity, Rotating Detonation Engine Simulations" In AIAA Scitech 2019 Forum, 2019

Ryan F. Johnson, Douglas A. Schwer, Andrew. D. Kercher, Andrew T. Corrigan, K. Kailasanath, and Ephraim Gutmark; "Flow Characteristics of a Recirculating Flameless Combustor Configuration"; 2018 AIAA Aerospace Sciences Meeting, 0136